

**Motor City Math**  
S4 Rational Expressions: Gap Drill

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Period: \_\_\_\_\_

**Question 1** S4.a

Simplify by factoring first:

$$\frac{x^2 - 4}{x^2 + 5x + 6}$$

Answer = \_\_\_\_\_

Show your work:

**Question 2** S5.d

Simplify by factoring first:

$$\frac{x^3 - 4x}{x^2 - x - 2}$$

Answer = \_\_\_\_\_

Show your work:

**Question 3**

S5.d

Simplify by factoring first:

$$\frac{x^2 + x}{x^2 - 1}$$

*Reminder: Factor completely before canceling.*

Answer = \_\_\_\_\_

Show your work:

**Question 4**

S4.a

Simplify by factoring first:

$$\frac{2x^2 - 8}{x^2 - 6x + 8}$$

Answer = \_\_\_\_\_

Show your work:

**Question 5** S5.d

Simplify by factoring first:

$$\frac{x^3 + 2x^2 - 3x}{x^2 + 4x + 3}$$

*Warning: If you try to cancel individual terms without factoring, you WILL get the wrong answer.*

Answer = \_\_\_\_\_

Show your work:

**Question 6** S4.b

Add. Write your answer as a single fraction in simplest form:

$$\frac{3}{x+2} + \frac{5}{x-1}$$

Answer = \_\_\_\_\_

Show your work:

**Question 7** S4.b

Subtract. Write your answer as a single fraction in simplest form:

$$\frac{4x}{x+1} - \frac{3}{x+4}$$

Answer = \_\_\_\_\_

Show your work:

**Question 8** S4.b

Add. Simplify your answer completely:

$$\frac{1}{2x^2} + \frac{3}{4x^3}$$

Answer = \_\_\_\_\_

Show your work:

**Question 9**

S4.b

Subtract. Factor and simplify completely:

$$\frac{x}{x-3} - \frac{2}{x+1}$$

Answer = \_\_\_\_\_

Show your work:

**Question 10**

S4.a

Expand and simplify:

$$3(2x + 1)^2$$

Answer = \_\_\_\_\_

Show your work:

**Question 11**

other

Simplify the difference quotient:

$$\frac{f(x+h) - f(x)}{h} \text{ where } f(x) = 3x^2 - 2x + 1$$

Show all expansion steps. Watch your coefficients.

Answer = \_\_\_\_\_

Show your work:

**Question 12**

S4.b

Subtract and simplify completely:

$$\frac{2x+3}{x+1} - \frac{x-2}{x+4}$$

Answer = \_\_\_\_\_

Show your work: